A List Game

You are playing the following simple game with a friend:

1. The first player picks a positive integer X.
2. The second player gives a list of k positive integers $Y_1, \ldots, Y_k$ such that $(Y_1 + 1)(Y_2 + 1) \cdots (Y_k + 1) = X$, and gets k points.

Write a program that plays the second player.

Shirley Chisholm examining political statistics in 1965 (from Wikipedia Commons).

Input

The input consists of a single integer X satisfying $10^3 \leq X \leq 10^9$, giving the number picked by the first player.

Output

Write a single integer k, giving the number of points obtained by the second player, assuming she plays as well as possible.

Sample Input 1

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65536
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Sample Output 1

```
16
```

Sample Input 2

```
127381
```

Sample Output 2

```
3
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